PHYS4055

Atomic Physics and Quantum Optics

Complete Worked Past Exam Solutions

2023 · 2024 · 2025

All 12 questions, including every subpart

2026 open-book examination preparation

What this book contains. All four questions from *each* of the three uploaded exam papers are solved step by step. The calculations include intermediate algebra, physical reasoning, level sketches and result checks. Although the papers require students to answer only three of four questions, this book covers *every* question.

How to use it: Read a past question first, attempt the derivation, then compare the worked steps. For a timed revision exercise, choose three questions per paper and work for two hours.

Important conventions and source caveats. The original PDFs in `source_material/past_exams` remain authoritative for diagrams and printed notation. When a problem leaves a speed convention or squeezed-state ordering unspecified, this book states a reasonable convention and also notes valid alternatives. In the 2024 sensor problem a dimensionally necessary κ factor is derived explicitly. These are independently prepared study solutions; no official past-exam marking scheme was provided.

Paper	Question themes
2023	Nonlinear optical cat states; H/He radial and spin structure; squeezed interferometry; Raman π-pulse sensing
2024	^3D spin–orbit splitting; Zeeman sensor; HOM and statistics; lossy cavity and dual-rail gates
2025	Hyperfine/Zeeman structure; Yb optical clock, cooling and Doppler limit; NOON-state beam splitter; parametric cavity squeezing and Bell keys

First prepared 13 November 2026. Source audit: `SOURCE_MAP.md`. Editable sections: `sections/{year}_q{n}.tex`.

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Semester 2 2023 examination

1 Question 1: A nonlinear optical mode: coherent states and Schrödinger cats

Original: PHYS4055 Semester 2 2023, Question 1, source paper page 2. All subparts answered.

Consider the Hamiltonian

$$H = \hbar[\omega a^\dagger a + \lambda (a^\dagger a)^p] = \hbar[\omega \hat{n} + \lambda \hat{n}^p].$$

Take p to be a *positive* integer and $\lambda \neq 0$ in the specified evolution time.

(a) Eigenstates and eigenvalues

Because H is a function of $\hat{n} = a^\dagger a$, all number states are eigenstates:

$$\boxed{H\,|n\rangle = E_n\,|n\rangle\,, \qquad E_n = \hbar(\omega n + \lambda n^p), \quad n = 0, 1, 2, \ldots}.$$

This is a number-dependent nonlinearity. The energy spacings are generally not constant.

(b) Evolve a coherent input in the interaction picture

Take $H_0 = \hbar\omega\hat{n}$; it commutes with $H_I = \hbar\lambda\hat{n}^p$. Write the initial coherent state as

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,.$$

Since $U_I(t) = \exp(-i\lambda t \hat{n}^p)$, the evolved state is

$$\boxed{|\psi_I(t)\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-i\lambda t n^p}\,|n\rangle}.$$

The nonlinear unitary changes phases but *not* photon-number probabilities.

(c) Recover the coherent-state cat for even p

For $p = 2, 4, \ldots$ and integer n, n^p is divisible by 4 if n is even, while $n^p \equiv 1 \pmod 4$ if n is odd. At the time $t = \pi/(2\lambda)$,

$$e^{-i\pi n^p/2} = \begin{cases} 1, & n \text{ even}, \\ -i, & n \text{ odd}. \end{cases}$$

The parity-dependent coefficient is

$$e^{-i\pi n^p/2} = \tfrac{1}{2}[(1-i) + (1+i)(-1)^n].$$

Substitute into the Fock expansion and use $(-1)^n\alpha^n = (-\alpha)^n$:

$$\boxed{|\psi_I\rangle = \frac{1}{2}[(1-i)\,|\alpha\rangle + (1+i)\,|-\alpha\rangle]}.$$

It is correctly normalised: the cross term between $|\alpha\rangle$ and $|-\alpha\rangle$ has purely imaginary coefficient, while $\langle\alpha|-\alpha\rangle = e^{-2|\alpha|^2}$ is real.

(d) Calculate all normal-ordered correlations

The diagonal photon-number probabilities remain

$$P_n = e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}.$$

For positive integer k, the normally ordered factorial moment is

$$\langle (a^\dagger)^k a^k \rangle = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)P_n = |\alpha|^{2k}.$$

As $\langle\hat{n}\rangle = |\alpha|^2$,

$$\boxed{g^{(k)}(0) = \frac{\langle (a^\dagger)^k a^k \rangle}{\langle \hat{n} \rangle^k} = 1 \qquad (k = 1, 2, \ldots,\ |\alpha| > 0).}$$

This is the usual *normalised zero-delay* intensity-correlation definition.

(e) Is the state classical?

Despite having *Poissonian photon statistics* and $g^{(k)}(0) = 1$, the state is a coherent superposition of distinguishable field amplitudes α and $-\alpha$, not a classical mixture. Its phase-space representation contains quantum interference fringes and, for nonzero well-resolved amplitudes, Wigner negativity; a positive classical mixture of coherent states cannot reproduce it. Photon counts alone fail to reveal these phase-sensitive coherences. For $\alpha = 0$ the state reduces to the vacuum (not a nontrivial cat).

> **Verification check.** The relative phase between Fock amplitudes changes, but $|e^{-i\lambda t n^p}|^2 = 1$, so every photon-number probability is unchanged.

2 Question 2: Hydrogen radial probabilities, helium spins and E1 lines

Original: PHYS4055 Semester 2 2023, Question 2, source paper page 3. All subparts answered.

The hydrogen $1s$ wavefunction is

$$\psi_{1s}(r) = \frac{e^{-r/a_0}}{\sqrt{\pi a_0^3}}, \qquad |\psi_{1s}|^2 = \frac{e^{-2r/a_0}}{\pi a_0^3}.$$

(a) Radius with maximum detection probability

The correct *radial probability distribution* is the probability in a spherical shell, $P(r)\mathrm{d}r = 4\pi r^2 |\psi|^2 \mathrm{d}r$:

$$P(r) = \frac{4r^2}{a_0^3} e^{-2r/a_0}.$$

Differentiating the log, $P'/P = 2/r - 2/a_0$. Apart from the zero endpoint, the maximum is

$$\boxed{r_{\text{most probable}} = a_0}.$$

Maximising $|\psi(r)|^2$ alone would wrongly give $r = 0$: it omits the increasing spherical-shell volume.

(b) Evaluate $\langle r \rangle$ and $\langle 1/r \rangle$

Using $\int_0^\infty r^n e^{-\beta r}\mathrm{d}r = n!/\beta^{n+1}$,

$$\langle r \rangle = \frac{4}{a_0^3}\int_0^\infty r^3 e^{-2r/a_0}\mathrm{d}r = \frac{4}{a_0^3}\frac{3!}{(2/a_0)^4} = \boxed{\frac{3a_0}{2}},$$
$$\left\langle \frac{1}{r} \right\rangle = \frac{4}{a_0^3}\int_0^\infty r e^{-2r/a_0}\mathrm{d}r = \frac{4}{a_0^3}\frac{1!}{(2/a_0)^2} = \boxed{\frac{1}{a_0}}.$$

Bohr's $n = 1$ orbit has fixed radius a_0; it agrees with the quantum *most probable* radius and $\langle 1/r \rangle = 1/a_0$, but not the mean radius $\langle r \rangle = 3a_0/2$.

(c) Schrödinger equation for helium

For a fixed nucleus $Z = 2$, a simplified non-relativistic two-electron Hamiltonian is

$$\boxed{H = -\frac{\hbar^2}{2m_e}(\nabla_1^2 + \nabla_2^2) - \frac{2e^2}{4\pi\epsilon_0}\left(\frac{1}{r_1} + \frac{1}{r_2}\right) + \frac{e^2}{4\pi\epsilon_0 r_{12}}.}$$

The three terms represent electron kinetic energy, attractions to the nucleus and mutual electron repulsion, respectively. Solve $H\Psi(\mathbf{r}_1, \mathbf{r}_2) = E\Psi$ with total fermion-exchange antisymmetry. Spin and relativistic interactions would be additional smaller corrections.

(d) All four two-electron spin states

Total S	Spin state	Exchange
$1,\ M_S = 1$	$\lvert\uparrow\uparrow\rangle$	symmetric
$1,\ M_S = 0$	$(\lvert\uparrow\downarrow\rangle + \lvert\downarrow\uparrow\rangle)/\sqrt{2}$	symmetric
$1,\ M_S = -1$	$\lvert\downarrow\downarrow\rangle$	symmetric
$0,\ M_S = 0$	$(\lvert\uparrow\downarrow\rangle - \lvert\downarrow\uparrow\rangle)/\sqrt{2}$	antisymmetric

The first three are the triplet, the last the singlet. Since total two-electron wavefunction is antisymmetric, a triplet needs an antisymmetric spatial part, while a singlet needs a symmetric spatial part.

(e) Physical origin and relative magnitude of hyperfine structure

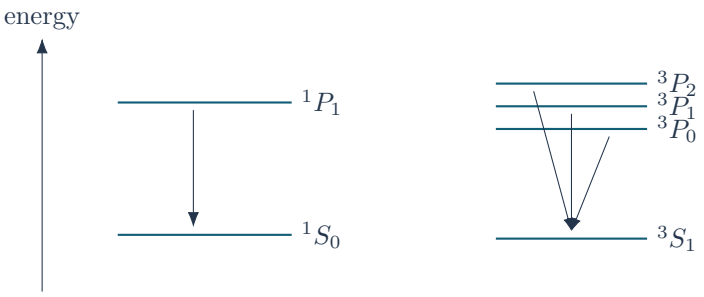
Fine structure primarily concerns electronic spin–orbit and relativistic effects. Hyperfine structure couples electronic angular momentum and fields to *nuclear* magnetic dipole/electric quadrupole moments. The nuclear magnetic moment is typically of nuclear-magneton scale, $\mu_N = e\hbar/(2m_p)$, much smaller than the electron Bohr magneton $\mu_B = e\hbar/(2m_e)$; this suppresses hyperfine interactions. For ^{4}He, whose nucleus has $I = 0$, magnetic-dipole hyperfine splitting is absent.

(f) Allowed E1 transitions among the stated terms

To leading order in LS coupling, E1 requires opposite orbital parity, $\Delta S = 0$, and $|\Delta J| \le 1$ with $J = 0 \not\to J' = 0$. Thus the allowed combinations are

$$\boxed{{}^1P_1 \longleftrightarrow {}^1S_0, \qquad {}^3P_{0,1,2} \longleftrightarrow {}^3S_1.}$$

The direction of spontaneous emission is from the higher to the lower member of each pair. Singlet–triplet lines are spin-forbidden in pure LS coupling but can gain weak intensity through spin–orbit mixing. A simple term diagram is:



Verification check. Radial expectations require the $4\pi r^2$ measure. For selection rules, check *spin*, *parity* and *angular momentum* independently.

3 Question 3: Squeezed-coherent input in a Mach–Zehnder interferometer

Original: PHYS4055 Semester 2 2023, Question 3, source paper page 4. All subparts answered.

We have one squeezed-coherent input mode a and one vacuum input mode v. For clarity, distinguish *field displacement* α from squeeze parameters $\mu = \cosh r,\ \nu = e^{2i\varphi_S}\sinh r$.

(a) Energy conservation at the first beam splitter

The stated beam splitter matrix is

$$M = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

The input column of field amplitudes is $\mathbf{E}_{\text{in}} = (E_{\text{in}}, E_v)^T$. Since

$$M^\dagger M = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = I,$$

we obtain

$$\boxed{|E_1|^2 + |E_2|^2 = \mathbf{E}_{\text{in}}^\dagger M^\dagger M \mathbf{E}_{\text{in}} = |E_{\text{in}}|^2 + |E_v|^2}.$$

For quantum modes the analogous statement is conservation of total photon number, because M is unitary.

(b) Will squeezing enhance the sensitivity?

Not *automatically*. Squeezing can lower noise in a selected quadrature, but the effect depends on which input port is squeezed, its squeezing angle, the operating phase and the measurement. In this question the *other input is vacuum* and the measured observable is the *output photon-number difference*. As the calculations below show, at $\phi = \pi/2$ the variance of the difference is the *mean photon number* of the occupied input. Thus this exact configuration does *not* give sub-shot-noise phase resolution at fixed total mean photon number through simple difference detection. Injecting appropriately squeezed vacuum in the unused port is a different well-known enhancement scheme.

(c) Derive the number-difference operator

The problem gives

$$a_A = c\,a + d\,v, \qquad a_B = d\,a + c\,v, \quad c = \tfrac{1}{2}(e^{i\phi}+1), \quad d = \tfrac{1}{2}(e^{i\phi}-1).$$

Compute the coefficients:

$$|c|^2 - |d|^2 = \cos\phi, \qquad c^* d - d^* c = i\sin\phi.$$

Expand $a_A^\dagger a_A - a_B^\dagger a_B$:

$$\boxed{D \equiv n_A - n_B = \cos\phi(n_a - n_v) + i\sin\phi(a^\dagger v - v^\dagger a)}.$$

(d) Signal for the squeezed-coherent input

Since the second mode is vacuum, $\langle n_v\rangle = 0$, $\langle v\rangle = 0$; all cross-mode expectation values factor. Hence

$$\boxed{\langle D\rangle = \cos\phi\, N}, \qquad N = \langle a^\dagger a\rangle.$$

For the conventional displaced-squeezed state $|\alpha,\xi\rangle = D(\alpha)S(\xi)\,|0\rangle$, we have

$$S^\dagger a S = \mu a - \nu a^\dagger, \qquad \boxed{N = |\alpha|^2 + |\nu|^2 = |\alpha|^2 + \sinh^2 r}.$$

Therefore

$$\boxed{\langle n_A - n_B\rangle = \cos\phi(|\alpha|^2 + |\nu|^2)}.$$

If the problem instead defines $|\alpha,\xi\rangle = S(\xi)D(\alpha)\,|0\rangle$, operator ordering changes the number expectation: $N = |\mu\alpha - \nu\alpha^*|^2 + |\nu|^2$. The subsequent noise formula in terms of N remains valid. *State the convention.*

(e) Number-difference variance at $\phi = \pi/2$

At the operating phase, $D = i(a^\dagger v - v^\dagger a)$ and $\langle D\rangle = 0$. Square it:

$$D^2 = -a^\dagger v a^\dagger v + a^\dagger v v^\dagger a + v^\dagger a a^\dagger v - v^\dagger a v^\dagger a.$$

In the vacuum v, $\langle v^\dagger v\rangle = 0$, $\langle v v^\dagger\rangle = 1$, $\langle v^2\rangle = \langle v^{\dagger 2}\rangle = 0$. Only the second term remains, so

$$\boxed{\mathrm{Var}(D)|_{\pi/2} = \langle D^2\rangle = \langle a^\dagger a\rangle = N = |\alpha|^2 + |\nu|^2}$$

for $D(\alpha)S(\xi)\,|0\rangle$.

(f) Minimum resolvable phase near $\pi/2$

For a small deviation ϵ, $\cos(\pi/2 + \epsilon) \simeq -\epsilon$. Thus the signal magnitude is $|\langle D\rangle| \simeq N|\epsilon|$, and the noise standard deviation is $\sqrt{N}$. Demanding signal-to-noise ratio 1 gives

$$1 \simeq \frac{N|\epsilon|}{\sqrt{N}}, \qquad \boxed{\delta\phi_{\min} \simeq \frac{1}{\sqrt{N}} = \frac{1}{\sqrt{|\alpha|^2 + \sinh^2 r}}}.$$

For M independent repetitions, the standard error scales as $1/\sqrt{MN}$. Squeezing can raise N if α is held fixed, but *at equal total photon number* this is still shot-noise scaling.

Verification check. The sensitivity must be computed using the *slope* at the dark operating point: the mean signal itself is zero exactly at $\phi = \pi/2$. The derivation does not assume a Gaussian input; only that the other port is vacuum.

4 Question 4: Raman transition, atomic Bloch equations and pulse-area sensing

Original: PHYS4055 Semester 2 2023, Question 4, source paper page 5–6. All subparts answered.

The question describes an effective two-photon Raman coupling between $|g\rangle, |e\rangle$ mediated by an auxiliary state $|a\rangle$. The effective Hamiltonian is already written in terms of a cavity photon and a control field; the off-resonant auxiliary state has been eliminated in the model.

(a) Remove the fast free oscillations

Write

$$\sigma_+ = |e\rangle\langle g|, \quad \sigma_- = |g\rangle\langle e|, \quad H_0 = \hbar\omega_a a^\dagger a + \frac{\hbar\omega_{eg}}{2}\sigma_z.$$

The interaction in the Schrödinger picture contains $\hbar g\mathcal{E}_c(t)a^\dagger\sigma_- e^{-i\omega_c t} + \text{h.c.}$. Under the free interaction picture,

$$a^\dagger\sigma_- \mapsto a^\dagger\sigma_- e^{i(\omega_a-\omega_{eg})t},$$

so the coupling acquires

$$e^{i(\omega_a-\omega_{eg}-\omega_c)t} = e^{-i\delta t}, \quad \delta = \omega_{eg} - (\omega_a - \omega_c).$$

A further rotating-frame phase transformation transfers this time-dependent phase to the cavity-number term. With a real control envelope, an equivalent form is

$$\boxed{H_I = -\hbar\delta\, a^\dagger a + \hbar g\mathcal{E}_c(t)(a^\dagger\sigma_- + a\sigma_+)}.$$

The sign of the displayed number term is set by the selected rotating-frame convention; the physics depends on the *relative* two-photon detuning.

(b) Derive the equations for z, w

For a strongly driven cavity mode replace $a \to \alpha$, $a^\dagger \to \alpha^*$, treating α as the effective coherent amplitude (including any detuning response). The atomic coupling is $H_{\text{at}} = \hbar g\mathcal{E}_c(t)(\alpha^*\sigma_- + \alpha\sigma_+)$. With

$$z = \langle\sigma_z\rangle, \quad w = \langle\sigma_+\rangle, \quad w^* = \langle\sigma_-\rangle,$$

and commutators $[\sigma_z, \sigma_\pm] = \pm 2\sigma_\pm$, $[\sigma_-, \sigma_+] = -\sigma_z$, the Heisenberg equation gives

$$\dot{z} = \frac{i}{\hbar}\langle[H_{\text{at}}, \sigma_z]\rangle = \boxed{2ig\mathcal{E}_c(t)(\alpha^* w^* - \alpha w)},$$

$$\dot{w} = \frac{i}{\hbar}\langle[H_{\text{at}}, \sigma_+]\rangle = \boxed{-ig\mathcal{E}_c(t)\alpha^* z}.$$

Its conjugate is $\dot{w}^* = ig\mathcal{E}_c(t)\alpha z$. The equations supplied in part (b) assume that the coupling phase can be taken time-independent in this rotating description.

(c) Pulse-area solution for complete inversion

The specified envelope is $\mathcal{E}_c(t) = \mathcal{E}_0 e^{-\gamma|t|}$. The *pulse area* is

$$\boxed{\mathcal{A} = \int_{-\infty}^{\infty}\mathcal{E}_c(t)\mathrm{d}t = \frac{2\mathcal{E}_0}{\gamma}}.$$

Define the new time coordinate $\tau(t) = \int_{-\infty}^{t}\mathcal{E}_c(t')\mathrm{d}t'$. Then $\mathrm{d}/\mathrm{d}t = \mathcal{E}_c(t)\mathrm{d}/\mathrm{d}\tau$. The equations reduce to

$$\frac{\mathrm{d}z}{\mathrm{d}\tau} = 2ig(\alpha^* w^* - \alpha w), \qquad \frac{\mathrm{d}w}{\mathrm{d}\tau} = -ig\alpha^* z.$$

Differentiate z once more and substitute the equations for w, w^*:

$$\frac{\mathrm{d}^2 z}{\mathrm{d}\tau^2} = 2ig[\alpha^*(ig\alpha z) - \alpha(-ig\alpha^* z)] = \boxed{-4g^2|\alpha|^2 z}.$$

The initial atom is in $|g\rangle$, so $z(-\infty) = -1$, $w(-\infty) = 0$. The resonant solution is

$$\boxed{z(t) = -\cos[2g|\alpha|\tau(t)]}, \quad P_e(t) = \frac{1+z(t)}{2} = \sin^2[g|\alpha|\tau(t)].$$

After the complete pulse,

$$\boxed{P_e(\infty) = \sin^2\left(\frac{2g|\alpha|\mathcal{E}_0}{\gamma}\right)}.$$

Complete population inversion requires

$$\boxed{\frac{4g|\alpha|\mathcal{E}_0}{\gamma} = (2m+1)\pi, \quad m = 0, 1, \ldots,}$$

equivalently $2g|\alpha|\mathcal{A} = (2m+1)\pi$. This is a π-*pulse* condition.

Where does detuning enter? These Bloch equations have constant-phase coupling and no separate atomic detuning term. For their exact area solution, the effective Raman resonance must be satisfied, $\delta = 0$, or the relevant detuning must already have been consistently absorbed into $\alpha(\delta)$. If finite two-photon detuning remains in the atomic Hamiltonian, the Bloch equations gain extra precession terms and the simple area result is not sufficient. For a *constant* drive of strength $\Omega_R = 2g|\alpha|\mathcal{E}_c$, one finds $\Omega_{\text{eff}} = \sqrt{\Omega_R^2 + \delta^2}$, with peak excitation probability $\Omega_R^2/\Omega_{\text{eff}}^2 < 1$ for $\delta \neq 0$. A pulsed detuned problem must be solved with its actual time-dependent Hamiltonian. The question supplies no explicit cavity response $\alpha(\delta)$, so a unique nonzero-detuning inversion condition cannot be inferred from the given data alone.

(d) Use it as a detuning sensor

Prepare $|g\rangle$, apply the control pulse, then measure P_e by state-selective fluorescence or another population readout. Scan the frequency difference $\omega_a - \omega_c$ and fit $P_e(\delta)$ to find the two-photon resonance $\delta = 0$. With a calibrated drive and detuning-dependent cavity amplitude $\alpha(\delta)$, the measured population provides a frequency-to-population transducer. For a locally linear sensor, choose a bias at a nonzero response slope: $\delta P_e \simeq P_e'(\delta_0)\,\delta\delta$. Noise, spontaneous emission, finite pulse bandwidth and fluctuations set the practical resolution.

Verification check. Check the initial condition $P_e(-\infty) = 0$, final inversion $P_e(\infty) = 1$ for an odd-π pulse, and population conservation $P_e + P_g = 1$. Avoid using a resonance-only pulse-area solution unchanged at finite detuning.

Semester 2 2024 examination

5 Question 1: Spin–orbit splitting of the ^3D term

Original: PHYS4055 Semester 2 2024, Question 1, source paper page 2. All subparts answered.

(a) Allowed values of J

The superscript $3 = 2S + 1$ gives $S = 1$; the letter D gives $L = 2$. Angular-momentum addition therefore allows

$$\boxed{J = |L - S|, \ldots, L + S = 1, 2, 3.}$$

(b) Derive the energy formula

With $\mathbf{J} = \mathbf{L} + \mathbf{S}$,

$$\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L}\cdot\mathbf{S}, \qquad H_{\mathrm{SO}} = A\mathbf{L}\cdot\mathbf{S}.$$

Using angular momenta in dimensionless $\hbar$ units, the eigenvalues satisfy

$$\boxed{\Delta E_J = \frac{A}{2}\{J(J+1) - L(L+1) - S(S+1)\}.}$$

If angular-momentum operators retain SI dimensions, A here absorbs the required $\hbar^{-2}$.

(c) Evaluate all three energies and draw a level pattern

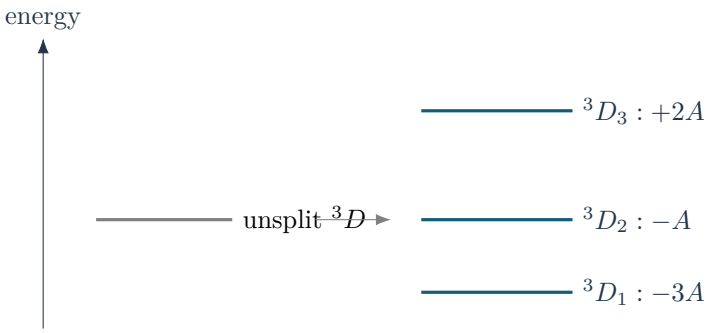
Since $L(L+1) = 6$ and $S(S+1) = 2$,

J	$J(J+1)$	ΔE_J	degeneracy $2J+1$
1	2	$\boxed{-3A}$	3
2	6	$\boxed{-A}$	5
3	12	$\boxed{+2A}$	7

For $A > 0$ the energy levels are ordered $J = 1, 2, 3$ from lowest to highest. The adjacent intervals are

$$\boxed{E_2 - E_1 = 2A, \qquad E_3 - E_2 = 3A}.$$

They are *not* equally spaced. A sketch:



The degeneracy-weighted centre of gravity is unchanged: $3(-3A) + 5(-A) + 7(2A) = 0$.

(d) Does spin–orbit coupling create more states?

No. The initial product basis has $(2L+1)(2S+1) = 5 \cdot 3 = 15$ states. The split multiplets have

$$\boxed{(2\cdot 1 + 1) + (2\cdot 2 + 1) + (2\cdot 3 + 1) = 3 + 5 + 7 = 15.}$$

The interaction redistributes energies without changing Hilbert-space dimension.

Verification check. The weighted-energy sum is zero and total degeneracy is conserved. The ordering assumes the stated $A > 0$.

6 Question 2: Zeeman-transition absorption as a magnetic-field sensor

Original: PHYS4055 Semester 2 2024, Question 2, source paper page 3. All subparts answered.

The problem supplies the steady-state Bloch inversion

$$\langle\sigma_z\rangle_{\mathrm{ss}} = -\frac{\gamma^2 + 4\Delta(B)^2}{\gamma^2 + 4\Delta(B)^2 + 8\Omega^2}, \qquad \Delta(B) = \kappa B.$$

Here Ω and γ are *frequencies*; the Hamiltonian's drive term must have energy units, i.e. effectively $\hbar\Omega\sigma_x$, if the printed shorthand is missing $\hbar$.

(a) Probability of the ground state

Since $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$, $P_e - P_g = \langle\sigma_z\rangle$ while $P_e + P_g = 1$. Therefore $P_g = (1 - \langle\sigma_z\rangle)/2$. Define $D = \gamma^2 + 4\kappa^2 B^2 + 8\Omega^2$, giving

$$P_g = \frac{1}{2}\left(1 + \frac{D - 8\Omega^2}{D}\right) = \boxed{1 - \frac{4\Omega^2}{\gamma^2 + 4\kappa^2 B^2 + 8\Omega^2}}.$$

On resonance $B = 0$, P_g is smallest; well away from resonance $P_g \to 1$.

(b) What is measured?

The AC absorption signal is assumed proportional to P_g. Calibrating the signal versus B gives a Lorentzian-shaped dip. For unknown perturbations near a known bias B_0,

$$P_g(B_0 + \delta B) \simeq P_g(B_0) + \left.\frac{\mathrm{d}P_g}{\mathrm{d}B}\right|_{B_0} \delta B.$$

Measuring the absorption change then estimates $\delta B \simeq \delta P_g / P_g'(B_0)$. The signal is *even* in B; its sign alone cannot distinguish $+B$ from $-B$. A known bias removes that ambiguity.

(c) Determine the optimal linear operating point

Put $C = \gamma^2 + 8\Omega^2$. Differentiate:

$$\frac{\mathrm{d}P_g}{\mathrm{d}B} = \frac{32\Omega^2\kappa^2 B}{(C + 4\kappa^2 B^2)^2}.$$

The sensitivity magnitude is maximised where the second derivative is zero:

$$\frac{\mathrm{d}^2 P_g}{\mathrm{d}B^2} = 32\Omega^2\kappa^2 \frac{C - 12\kappa^2 B^2}{(C + 4\kappa^2 B^2)^3} = 0.$$

Thus $12\kappa^2 B_0^2 = C$, so

$$\boxed{B_0 = \pm\frac{\sqrt{\gamma^2 + 8\Omega^2}}{2\sqrt{3}\,|\kappa|}}.$$

Dimensional check. The numerator has units of frequency, so the expression for B_0 must include division by κ (frequency per magnetic field). If the source print or text extraction omits this factor, the formula as printed cannot be dimensionally correct. The result above follows unambiguously from the supplied $\Delta(B) = \kappa B$.

Verification check. The slope vanishes at $B = 0$ and far off resonance. The inflection points at $\pm B_0$ produce maximum local linear response.

7 Question 3: HOM bunching, energy conservation and photon statistics

Original: PHYS4055 Semester 2 2024, Question 3, source paper page 4. All subparts answered.

The balanced beam-splitter convention in the question is

$$c = \frac{a + b}{\sqrt{2}}, \qquad d = \frac{a - b}{\sqrt{2}}, \quad \Longrightarrow \quad a^\dagger = \frac{c^\dagger + d^\dagger}{\sqrt{2}}, \quad b^\dagger = \frac{c^\dagger - d^\dagger}{\sqrt{2}}.$$

(a) Output for $|1,1\rangle$

Generate the input from vacuum:

$$|1,1\rangle_{ab} = a^\dagger b^\dagger |0,0\rangle .$$

On substitution,

$$a^\dagger b^\dagger |0\rangle = \frac{1}{2}[(c^\dagger)^2 - (d^\dagger)^2]|0\rangle = \boxed{\frac{|2,0\rangle_{cd} - |0,2\rangle_{cd}}{\sqrt{2}}}.$$

The $|1,1\rangle_{cd}$ amplitude is exactly zero: two indistinguishable photons bunch.

(b) Photon-number conservation for arbitrary inputs

Expand both output number operators:

$$c^\dagger c = \tfrac{1}{2}(a^\dagger a + a^\dagger b + b^\dagger a + b^\dagger b),$$
$$d^\dagger d = \tfrac{1}{2}(a^\dagger a - a^\dagger b - b^\dagger a + b^\dagger b).$$

All cross terms cancel, giving the operator identity

$$\boxed{c^\dagger c + d^\dagger d = a^\dagger a + b^\dagger b}.$$

With equal carrier frequencies this also conserves optical energy.

(c) Number variance for adjacent Fock states

For $|\psi\rangle = \alpha|n\rangle + \beta|n+1\rangle$, let $q = |\beta|^2$. Then

$$\langle n\rangle = (1-q)n + q(n+1) = n + q,$$
$$\langle n^2\rangle = (1-q)n^2 + q(n+1)^2 = n^2 + q(2n+1),$$

so

$$\boxed{\mathrm{Var}(n) = q - q^2 = q(1-q)}.$$

Compare with the mean: for a non-vacuum state $\mathrm{Var}(n) < n + q = \langle n\rangle$. The only equality in this family occurs for the trivial vacuum $n = q = 0$.

(d) Physical significance

Sub-Poissonian number fluctuations cannot arise from a mixture of classical coherent fields with a positive Glauber–Sudarshan P distribution; they are an operational signature of nonclassical light. A conventional coherent state instead has $\mathrm{Var}(n) = \langle n\rangle$.

> **Verification check.** In part (a), the two probabilities are 1/2 and 1/2, so they sum to 1. In part (c), the variance depends only on the probability q, not the base integer n.

8 Question 4: Two-port cavity Langevin equation and photonic qubit encoding

Original: PHYS4055 Semester 2 2024, Question 4, source paper page 5. All subparts answered.

Given the cavity damping rate $\gamma = \gamma_1 + \gamma_2$,

$$\dot{a} = -\frac{\gamma}{2}a + \sqrt{\gamma_1}A_{\mathrm{in},1} + \sqrt{\gamma_2}A_{\mathrm{in},2}.$$

(a) Fourier-domain intracavity field

Take $\tilde{a}(\omega) = \int e^{-i\omega t}a(t)\mathrm{d}t$, so $\widetilde{\dot{a}} = i\omega\tilde{a}$. Collecting terms,

$$\left(\frac{\gamma}{2} + i\omega\right)\tilde{a} = \sqrt{\gamma_1}\tilde{A}_{\mathrm{in},1} + \sqrt{\gamma_2}\tilde{A}_{\mathrm{in},2},$$

$$\boxed{\tilde{a} = \frac{\sqrt{\gamma_1}\tilde{A}_{\mathrm{in},1} + \sqrt{\gamma_2}\tilde{A}_{\mathrm{in},2}}{\gamma/2 + i\omega}}.$$

(b) Field reflected from the first port

Use the specified boundary condition $\tilde{A}_{\text{out},1} = \sqrt{\gamma_1}\tilde{a} - \tilde{A}_{\text{in},1}$. After substituting,

$$\boxed{\tilde{A}_{\text{out},1} = \frac{\gamma_1 - \gamma/2 - i\omega}{\gamma/2 + i\omega}\tilde{A}_{\text{in},1} + \frac{\sqrt{\gamma_1\gamma_2}}{\gamma/2 + i\omega}\tilde{A}_{\text{in},2}.}$$

The second input is the internal loss reservoir and is required in the *operator* output, even if its mean coherent amplitude is zero.

(c) Critical coupling at zero Fourier frequency

For $\gamma_1 = \gamma_2$, we have $\gamma_1 = \gamma/2$. Therefore the *coherent reflection coefficient* at $\omega = 0$ is

$$\boxed{r_1(0) = \frac{\gamma_1 - \gamma/2}{\gamma/2} = 0}.$$

A coherent field incident at port 1 then has zero mean resonant reflected amplitude and, ideally, no reflected coherent intensity. Energy is lost through port 2 (the internal-loss channel); unavoidable reservoir vacuum fluctuations remain in the output operator.

(d) Two dual-rail encodings

A dual-rail qubit places one photon in exactly one of two distinguishable optical modes.

- **Spatial path**: $|0_L\rangle = |1_a 0_b\rangle$, $|1_L\rangle = |0_a 1_b\rangle$. Natural for chip-based interferometers with spatially separate waveguides.
- **Polarisation modes**: $|0_L\rangle = |H\rangle$, $|1_L\rangle = |V\rangle$, i.e. occupancy of the H/V polarisation modes. Efficient in a single fibre/path where maintaining two spatial arms is inconvenient.

Path encoding is often robust to uniform polarisation rotations but sensitive to relative path phase; polarisation encoding avoids distinct paths but can suffer birefringence and polarisation drift.

(e) Implement a Hadamard physically

For the logical basis, the Hadamard is

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

In **path encoding**, a balanced beam splitter with the appropriate phase convention implements H (up to phase shifters). In **polarisation encoding**, a half-wave plate with its fast axis at 22.5° to horizontal implements $|H\rangle \mapsto (|H\rangle + |V\rangle)/\sqrt{2}$, $|V\rangle \mapsto (|H\rangle - |V\rangle)/\sqrt{2}$, up to overall/relative phases set by conventions.

Verification check. At equal rates, the two-port cavity must still preserve the combined input/output commutation relations; discarding port 2 at the operator level would violate quantum noise balance.

Semester 2 2025 examination

9 Question 1: Hyperfine structure, weak/strong Zeeman splitting and quadrupole rules

Original: PHYS4055 Semester 2 2025, Question 1, source paper page 2. All subparts answered.

The specified electronic term $^2P_{1/2}$ has $L = 1, S = 1/2, J = 1/2$, coupled to $I = 5/2$.

(a) Allowed F levels

$\mathbf{F} = \mathbf{I} + \mathbf{J}$ gives $F = |I - J|, \ldots, I + J$, therefore $\boxed{F = 2,\ 3}$.

(b) Derive the magnetic hyperfine energies

With $H_{\text{hfs}} = A\,\mathbf{I}\cdot\mathbf{J}/\hbar^2$, and $\mathbf{F}^2 = \mathbf{I}^2 + \mathbf{J}^2 + 2\mathbf{I}\cdot\mathbf{J}$, take expectation values:

$$\boxed{E_F = \frac{A}{2}[F(F+1) - I(I+1) - J(J+1)]}.$$

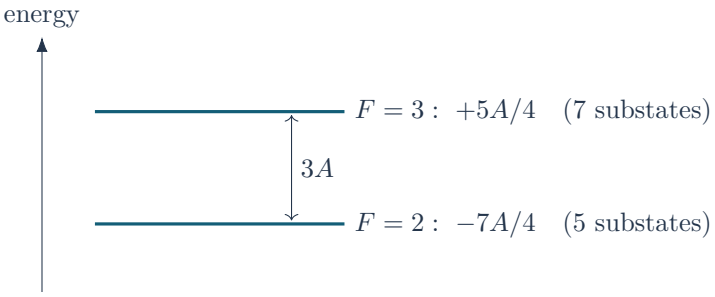
The convention absorbs $\hbar^{-2}$ into the Hamiltonian when angular momenta are written dimensionlessly.

(c) Find the splitting and sketch the diagram

Substitute $I(I+1) = 35/4$, $J(J+1) = 3/4$:

$$E_{F=2} = \frac{A}{2}\left(6 - \frac{35}{4} - \frac{3}{4}\right) = \boxed{-\frac{7A}{4}},$$
$$E_{F=3} = \frac{A}{2}\left(12 - \frac{35}{4} - \frac{3}{4}\right) = \boxed{\frac{5A}{4}}.$$

The signed difference is $E_3 - E_2 = 3A$; the physical frequency separation is $\boxed{\Delta\nu = 3|A|/h}$. For $A > 0$, the $F = 3$ multiplet lies higher.



(d.i–ii) Weak magnetic field

For $\mu_B|B| \ll |A|$ (up to numerical factors), F, m_F remain approximately good quantum numbers:

$$E_{Fm_F} \simeq E_F + \mu_B g_F m_F B, \quad g_F \simeq g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}.$$

For $^2P_{1/2}$, $g_J = 2/3$; therefore

$$\boxed{g_2 = -\frac{1}{9}, \quad g_3 = +\frac{1}{9}.}$$

The $F = 2$ level becomes $m_F = -2, -1, 0, 1, 2$ (5 lines) and $F = 3$ becomes $-3, \ldots, 3$ (7 lines). Thus generically there are $\boxed{12}$ distinct weak-field substates. Each multiplet fans into equally separated lines in m_F, but the two multiplets have opposite slopes.

(d.iii) Strong magnetic field

When $\mu_B g_J B \gg |A|$, the preferred basis is $|Jm_J, Im_I\rangle$, with two electronic branches $m_J = \pm 1/2$, each containing six m_I values:

$$E_{m_J m_I} \simeq \mu_B g_J m_J B + A\, m_J m_I \quad \text{(small nuclear Zeeman term neglected)}.$$

The level pattern looks like two steep approximately linear electronic fans, each split by six hyperfine offsets; it is not appropriate to extrapolate fixed-F lines into this regime.

(e) Are higher nuclear multipoles allowed?

The nuclear spin $I = 5/2$ *can* support a nuclear electric quadrupole (rank 2), but its diagonal hyperfine contribution also requires a rank-2 electronic tensor matrix element $\langle J\|T^{(2)}\|J\rangle$. By Wigner–Eckart the $3j$ triangle condition is $2 \le J + J = 1$, which fails for $J = 1/2$. Thus the $\boxed{\text{electric-quadrupole hyperfine term vanishes for this level}}$, as do higher ranks $k > 1$ at first order within the fixed $J = 1/2$ manifold.

Verification check. Total state count: $5 + 7 = 12 = (2I + 1)(2J + 1)$. The existence of a nuclear quadrupole alone does not guarantee a nonzero electronic quadrupole interaction.

10 Question 2: Ytterbium level diagram, slowing and laser cooling

Original: PHYS4055 Semester 2 2025, Question 2, source paper page 3. All subparts answered.

The source paper supplies a Yb level diagram, including $6s^2\,{}^1S_0$, $6s6p$ P and $5d6s$ D terms. Distinguish strictly LS-allowed E1 transitions from weak spin-mixed intercombination lines.

(a) Identify E1 arrows

Electric dipole (E1) has odd parity and angular rank 1. In a good LS-coupling approximation, require a parity change, $\Delta S = 0$, and $\Delta J = 0, \pm 1$ excluding $0 \leftrightarrow 0$. For the terms visible in the supplied diagram,

Level pair	Strict LS E1?
${}^1P_1 \leftrightarrow {}^1S_0$	yes
${}^1P_1 \leftrightarrow {}^1D_2$	yes
${}^3P_1 \leftrightarrow {}^3D_1$	yes
${}^3P_1 \leftrightarrow {}^3D_2$	yes
${}^3P_1 \leftrightarrow {}^3D_3$	no, $\Delta J = 2$
${}^3P_1 \leftrightarrow {}^1S_0$	spin-forbidden in pure LS coupling

With spin–orbit mixing the ${}^3P_1 \leftrightarrow {}^1S_0$ line becomes a weak *intercombination* E1 transition. A $J = 0 \leftrightarrow 0$ line remains forbidden for a rank-1 operator even with state mixing. If additional P_J levels are present in the original figure, apply the same rules to each. See the original level diagram in the paper and the corresponding page preview retained with the source extractions.

(b) Clock transition

A clock favours a spectrally narrow, long-lived transition over a bright, rapidly decaying one. Among the P terms visible in the supplied diagram, one candidate is the much narrower intercombination transition

$$\boxed{6s^2\,{}^1S_0 \longleftrightarrow 6s6p\,{}^3P_1 \quad (\sim 556\text{ nm})}.$$

It is spin-forbidden in pure LS coupling, but spin–orbit mixing makes it weakly allowed. This is narrower than the bright ${}^1S_0 \to {}^1P_1$ cooling line and is a possible frequency reference. The *standard highest-precision* optical Yb clock instead uses ${}^1S_0 \leftrightarrow {}^3P_0$ near 578 nm, a nominally E1-forbidden $J = 0 \to 0$ line, in isotopes where it can be interrogated. Choose that state if it appears in the original level diagram; either answer must explain why the chosen excited state is long-lived.

(c) Laser-cooling transition

Use the strong, nearly cycling

$$\boxed{6s^2\,{}^1S_0 \longleftrightarrow 6s6p\,{}^1P_1 \quad (\lambda \simeq 399\text{ nm})}.$$

Its large spontaneous-emission rate allows many momentum kicks. The narrower ${}^1S_0 \to {}^3P_1$ line near 556 nm is useful for subsequent cooling.

(d.i) Why the Zeeman slower needs a magnet

As atoms slow, their Doppler shift changes, tending to move the laser away from resonance. A position-dependent magnetic field Zeeman-shifts the transition and compensates the changing Doppler shift, maintaining efficient scattering.

(d.ii–iii) Stopping distance and number of photons

The data are $M = 1.73 \times 10^{-25}$ kg, $T = 300$ K, $\lambda = 399$ nm, and excited-state lifetime $\tau = 5.5$ ns. The paper does not specify which characteristic thermal beam speed to use. Taking the 3D rms speed as an explicit convention,

$$v_0 = \sqrt{\frac{3k_BT}{M}} = \boxed{268.0\ \mathrm{m\,s^{-1}}}.$$

At saturation a two-level transition has maximum photon-scattering rate $\Gamma/2$ where $\Gamma = 1/\tau$. Each photon transfers mean directed momentum h/λ, so

$$F_{\max} = \frac{h}{2\lambda\tau}, \quad a_{\max} = \frac{h}{2M\lambda\tau}.$$

At *half* the maximum deceleration,

$$a = \frac{a_{\max}}{2} = \frac{h}{4M\lambda\tau} = \boxed{4.36 \times 10^5\ \mathrm{m\,s^{-2}}}.$$

Using $0 = v_0^2 - 2aL$,

$$\boxed{L = \frac{v_0^2}{2a} = 0.0823\ \mathrm{m} = 8.23\ \mathrm{cm}}.$$

Momentum balance requires

$$\boxed{N_\gamma = \frac{Mv_0}{h/\lambda} = \frac{Mv_0\lambda}{h} \simeq 2.79 \times 10^4}.$$

If instead one uses the Maxwell most-probable speed $\sqrt{2k_BT/M} = 218.8\ \mathrm{m\,s^{-1}}$, the corresponding results are $L = 5.49$ cm and $N_\gamma \simeq 2.28 \times 10^4$. State the chosen speed definition in the exam.

(e.i) Red-detuned optical molasses

An atom moving into a counter-propagating red-detuned beam sees its light shifted toward resonance and preferentially absorbs photons that oppose its motion. The average force near rest has the form $\mathbf{F} \simeq -\beta\mathbf{v}$.

(e.ii) Doppler limit

For a simple two-level transition,

$$T_D = \frac{\hbar\Gamma}{2k_B} = \frac{h}{4\pi k_B\tau} = \boxed{6.94 \times 10^{-4}\ \mathrm{K} = 0.694\ \mathrm{mK}}.$$

Verification check. Cooling requires a bright, rapid cycling transition; clocks require a narrow, long-lived one. Check the factor of one-half in the stated deceleration and the units of acceleration, length, temperature and photon count.

11 Question 3: Two-photon NOON input and beam-splitter output probabilities

Original: PHYS4055 Semester 2 2025, Question 3, source paper page 4. All subparts answered.

With the paper's balanced beam splitter $c = (a+b)/\sqrt{2}$, $d = (a-b)/\sqrt{2}$, the inverse relations for creation operators are

$$a^\dagger = \frac{c^\dagger + d^\dagger}{\sqrt{2}}, \qquad b^\dagger = \frac{c^\dagger - d^\dagger}{\sqrt{2}}.$$

(a) Transform the plus NOON input

Using $|2,0\rangle_{ab} = (a^\dagger)^2|0\rangle/\sqrt{2}$,

$$|2,0\rangle \longrightarrow \frac{1}{2}|2,0\rangle_{cd} + \frac{1}{\sqrt{2}}|1,1\rangle_{cd} + \frac{1}{2}|0,2\rangle_{cd}.$$

Similarly

$$|0,2\rangle \longrightarrow \frac{1}{2}|2,0\rangle_{cd} - \frac{1}{\sqrt{2}}|1,1\rangle_{cd} + \frac{1}{2}|0,2\rangle_{cd}.$$

The initial state is $|\Psi_{\text{in}}\rangle = (|2,0\rangle + |0,2\rangle)/\sqrt{2}$. Add the two output *amplitude vectors*:

$$\boxed{|\Psi_{\text{out}}\rangle = \frac{|2,0\rangle_{cd} + |0,2\rangle_{cd}}{\sqrt{2}}}.$$

(b) Detection probabilities

Read probabilities from squared moduli:

$$\boxed{P_{20} = \frac{1}{2}, \qquad P_{02} = \frac{1}{2}, \qquad P_{11} = 0}.$$

They add to 1.

(c) Physical interpretation

The two input components are *coherent* number-state amplitudes, so the histories leading to one photon in each output port interfere destructively. The input is a two-photon path-entangled NOON state. The output has perfect two-photon bunching, but here it originates from the plus-NOON superposition: this is not the same initial state as the ordinary two-single-photon HOM experiment. The *minus* NOON state in the same convention would instead map to $|1,1\rangle_{cd}$.

(d) An alternative superposition

For $|\Phi_{\text{in}}\rangle = (|1,1\rangle + |2,0\rangle)/\sqrt{2}$, first transform the $|1,1\rangle$ part:

$$a^\dagger b^\dagger|0\rangle = \frac{(c^\dagger)^2 - (d^\dagger)^2}{2}|0\rangle = \frac{|2,0\rangle - |0,2\rangle}{\sqrt{2}}.$$

This contains *no* $|1,1\rangle_{cd}$. The $|2,0\rangle$ input contributes $|1,1\rangle$ with amplitude $1/\sqrt{2}$. Including the initial superposition coefficient $1/\sqrt{2}$ yields

$$\boxed{\mathcal{A}_{11} = \frac{1}{2}, \qquad P_{11} = |\mathcal{A}_{11}|^2 = \frac{1}{4}}.$$

Verification check. Never square each path probability before adding coherent amplitudes; the cancellation in part (a) would be lost. Account for $\sqrt{2!}$ when normalising the two-photon states.

12 Question 4: Parametric cavity squeezing and entanglement-based random bits

Original: PHYS4055 Semester 2 2025, Question 4, source paper page 5. All subparts answered.

Start from the supplied quantum Langevin model,

$$\dot{a} = -\frac{\gamma}{2}a + \sqrt{\gamma}A_{\text{in}}, \quad H_i = \frac{i\hbar\chi}{2}(a^2 - a^{\dagger 2}).$$

(a) Derive the modified quantum Langevin equation

The Heisenberg interaction term is $\dot{a}|_{H_i} = (i/\hbar)[H_i, a]$. Using $[a^2, a] = 0$, $[a^{\dagger 2}, a] = -2a^\dagger$, we find

$$\frac{i}{\hbar}[H_i, a] = \frac{i}{\hbar}\frac{i\hbar\chi}{2}(2a^\dagger) = -\chi a^\dagger.$$

Thus

$$\boxed{\dot{a} = -\chi a^\dagger - \frac{\gamma}{2}a + \sqrt{\gamma}A_{\text{in}}}.$$

(b) Fourier-domain amplitude quadrature

Taking the adjoint (real χ), and defining $X_+ = a + a^\dagger$, $X_{+,\mathrm{in}} = A_{\mathrm{in}} + A_{\mathrm{in}}^\dagger$,

$$\dot{X}_+ = -\left(\chi + \frac{\gamma}{2}\right) X_+ + \sqrt{\gamma} X_{+,\mathrm{in}}.$$

Adopt the Fourier convention $\tilde{X}(\omega) = \int e^{-i\omega t} X(t) \mathrm{d}t$; then $\widetilde{\dot{X}} = i\omega \tilde{X}$, giving

$$\boxed{\tilde{X}_+(\omega) = \frac{\sqrt{\gamma}}{\gamma/2 + \chi + i\omega} \tilde{X}_{+,\mathrm{in}}(\omega)}.$$

(c) Input–output relation

For the convention $A_{\mathrm{out}} = \sqrt{\gamma}\, a - A_{\mathrm{in}}$,

$$\tilde{X}_{+,\mathrm{out}} = \sqrt{\gamma}\, \tilde{X}_+ - \tilde{X}_{+,\mathrm{in}} = \left[\frac{\gamma}{\gamma/2 + \chi + i\omega} - 1\right] \tilde{X}_{+,\mathrm{in}}.$$

Therefore

$$\boxed{\tilde{X}_{+,\mathrm{out}} = \frac{\gamma/2 - \chi - i\omega}{\gamma/2 + \chi + i\omega} \tilde{X}_{+,\mathrm{in}}}.$$

This agrees with the expression requested in the exam.

(d) When does squeezing occur?

For vacuum input with unit quadrature noise spectrum, the output *fluctuation spectrum* is

$$V_+(\omega) = \left|\frac{\gamma/2 - \chi - i\omega}{\gamma/2 + \chi + i\omega}\right|^2 = \boxed{\frac{(\gamma/2 - \chi)^2 + \omega^2}{(\gamma/2 + \chi)^2 + \omega^2}}.$$

For $\boxed{0 < \chi < \gamma/2}$, the steady system is below parametric threshold and $V_+(\omega) < 1$ for finite ω; the largest reduction occurs near $\omega = 0$. The orthogonal quadrature is anti-squeezed.

Exam notation issue. The prompt's shorthand $V_+ = |\langle \tilde{X}_{+,\mathrm{out}} \rangle|^2$ is not the usual *noise spectral variance*: for a vacuum input, the mean is zero even though its fluctuations are nonzero. The physically relevant squeezing test uses the fluctuation autocorrelation / power spectrum as above.

(e) Random and secret bits from Bell pairs

For $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, both parties measuring in the same Z basis obtain 00 or 11, each with probability 1/2. Their local marginal states are maximally mixed: $\rho_A = \rho_B = I/2$, hence every local bit is unpredictable. The outcomes nevertheless match perfectly. If the pairs are *verified* as high-fidelity entangled states and Eve is uncorrelated with them, monogamy limits what Eve can know. In practice secrecy is not guaranteed just by *assuming* Bell pairs: Alice and Bob should estimate disturbance, authenticate their discussion and perform error correction/privacy amplification as needed.

Verification check. The output squeezing factor approaches 1 at large $|\omega|$ and is below 1 at $\omega = 0$ for positive χ below threshold.

Rapid method index

Question	Most reusable starting method
2023 Q1	Diagonalise $f(\hat{n})$ in Fock basis; expand a coherent state; inspect parity-dependent phases
2023 Q2	Use $4\pi r^2\|\psi\|^2$; shell probability and gamma integrals; enforce fermion antisymmetry/E1 rules
2023 Q3	Show optical matrix unitarity; expand output number difference; compute noise with vacuum input; divide by slope
2023 Q4	Enter interaction picture; use spin commutators; define integrated pulse time; solve Bloch oscillator
2024 Q1	Couple $L, S \to J$; use $2\mathbf{L}\cdot\mathbf{S} = J^2 - L^2 - S^2$
2024 Q2	Convert $\langle\sigma_z\rangle$ to P_g, differentiate twice; apply dimensions to bias field
2024 Q3	Transform creation operators; use normalised Fock states; compute mean and variance
2024 Q4	Fourier transform Langevin equation; apply input–output relation; keep the loss reservoir
2025 Q1	Couple $I, J \to F$; calculate E_F; switch from coupled to uncoupled basis at high B
2025 Q2	Test E1 parity/$\Delta J/\Delta S$; scattering $h/(2\lambda\tau)$; kinematics; Doppler limit $\hbar/(2k_B\tau)$
2025 Q3	Expand two-photon creation operators and sum *amplitudes* before probabilities
2025 Q4	Heisenberg commutator; form X_+; Fourier solve and input–output; noise spectrum

Exam method. Start by writing the relevant model, operator definitions and basis. Evaluate symbolically before numerical substitution. Check probability and normalisation, limiting cases, angular-momentum degeneracy, selection-rule triangle/parity conditions and units. Explain the physical observable in words.